What prostate HM-MRI data can and cannot determine: a simulation audit of physics-informed VAE uncertainty

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Impact

In simulated prostate HM-MRI at SNR 20, tissue fractions were recoverable but compartmental diffusivity and T2 were not. A variational network's intervals were far too narrow; a noise-matched likelihood with full covariance helped without guaranteeing coverage.

Synopsis

Motivation: Epithelium fraction from hybrid multidimensional MRI can separate prostate cancer from benign tissue; a variational neural network could add per-voxel uncertainty.

Goal(s): Test, in simulation, what the data determine and whether that uncertainty holds.

Approach: We trained physics-informed networks on simulated signals, scored each compartment parameter, and compared 95% intervals with the information limit.

Results: No method estimated compartmental diffusivity or T2 better than a fixed population average. Initially, 95% intervals contained the truth in 5–62% of voxels; with a noise-matched loss, full-covariance tissue-fraction intervals covered 86–91% at SNR 20, but not across SNR.

Category: Primary: 703 Diffusion > Diffusion Modeling; Secondary: 311 Analysis Methods > Generative Models

Introduction

Hybrid multidimensional MRI (HM-MRI) fits a three-compartment (epithelium, stroma, lumen) diffusion-relaxation model to a b-value × echo-time grid[1,2]; epithelium fraction separates cancer from benign tissue[2] and agrees with histology[3]. A physics-informed autoencoder (PIA) fits it far faster than nonlinear least squares (NLLS)[4] but returns point estimates. Variational HM-MRI has been prototyped[5], and full-covariance or flow-based posteriors have been validated for CEST/MT and microstructure[6,7], but variational posteriors tend to be too narrow[8]. In simulation, we test what the data determine, whether a three-compartment physics-informed variational autoencoder (PIVA) is calibrated, and why not.

Methods

Simulation. $S = 1000 \sum_c v_c e^{-bD_c} e^{-\Delta TE/T2_c}$ was sampled at four b-values (0–1500 s/mm²) × four ΔTE (0–143 ms after minimum TE), as in ref. 4. With relative TE and known S_0, v_c are signal fractions at minimum TE, and fitting is easier than in vivo. Patient and voxel parameters were drawn hierarchically inside a prior box (Fig. 1). Rician noise: training σ ~ U(0.01, 0.08); test σ = 0.05 unless stated (SNR 20 at b = ΔTE = 0; 0.35 at b = 1500, longest TE). Test: 2000 voxels, 200 unseen patients.

Models. NLLS: single-start bounded least squares (1999/2000 fits ended on a bound). PIA: reimplementation of ref. 4, shortened training. PIVA-Diagonal: the PIA encoder with a diagonal Gaussian latent (Fig. 1); loss = squared error to the noise-free signal + 0.1·KL[q‖N(0, I)], a Gaussian likelihood with SD 0.9 signal units against noise SD 10–80. PIVA-CholeskyILR adds a Cholesky-factored full-covariance posterior[6,9] and log-ratio coordinates for v[10]. Calibrated-decoder retraining[11] of both posteriors (same noise-free target) set the likelihood SD to 50 (the test noise) or to each training voxel's own noise SD; the network never receives σ. Three initialisation seeds per model.

Evaluation. Per-compartment R^2 and MAE against a constant prior-centre predictor; coverage of mean ± 1.96 SD; width clipped to the prior box. Post hoc, one scale factor per parameter (σ-scaling[12]) was fit on a separate cohort. Widths were compared with a fixed interval (central 95% of the training distribution) and the information limit (Bayesian Cramér-Rao bound, BCRB; Rician Fisher information, population prior).

Results

Accuracy (Fig. 2). Per compartment, every D and T2 R^2 was negative and the constant had the lowest MAE; pooled over compartments it scored R^2 0.993 (D) and 0.975 (T2), above every model. By the BCRB, the data shrink the population prior SD by under 13% for D and T2 ($R^2 \leq 0.24$), and without the prior the CRB 95% width is 1.3–12 times the box. Fractions were estimated, with PIA best or tied: networks reached R^2 about 0.9 (lumen), 0.46–0.63 (epithelium) and 0.30–0.48 (stroma), against NLLS 0.75, 0.28 and −0.25.

Original loss (Fig. 3). Raw coverage was 0.05–0.33 (Diagonal) and 0.13–0.62 (CholeskyILR). For voxels within 0.05 of the benign and cancer epithelium values (0.23, 0.49: volume fractions[2], treated as signal fractions), CholeskyILR estimated 0.26 and 0.40, halving the contrast (shrinkage[13]), and covered 15% of cancer-like truths. σ-scaling reached 0.93–0.96 coverage, but D and T2 intervals (68–90% of the box) were wider than the fixed interval (45–69%).

Noise-matched likelihoods (Fig. 4). With SD 50, CholeskyILR fraction intervals covered 0.86–0.91 (Diagonal: 0.71–0.91) at σ = 0.05 without post-hoc scaling, at widths near the information limit, and partly restored contrast (0.29 and 0.47); D and T2 intervals spanned 88–99% of the box (posterior collapse[14]). Fixed-SD widths ignored σ: coverage fell from 0.94–0.98 (σ = 0.01) to 0.81–0.84 (σ = 0.08). Per-voxel SDs kept widths within 7 points of the limit for $\sigma \geq 0.03$, but at σ = 0.01 point errors stayed about three times the limit, so epithelium and stroma coverage fell to 0.64–0.67. Epithelium coverage was 0.93–0.96 near the benign and cancer values but 0.71 below 0.18, under both likelihoods. SD-50 intervals excluded the benign value in 49% of cancer-like voxels.

Discussion

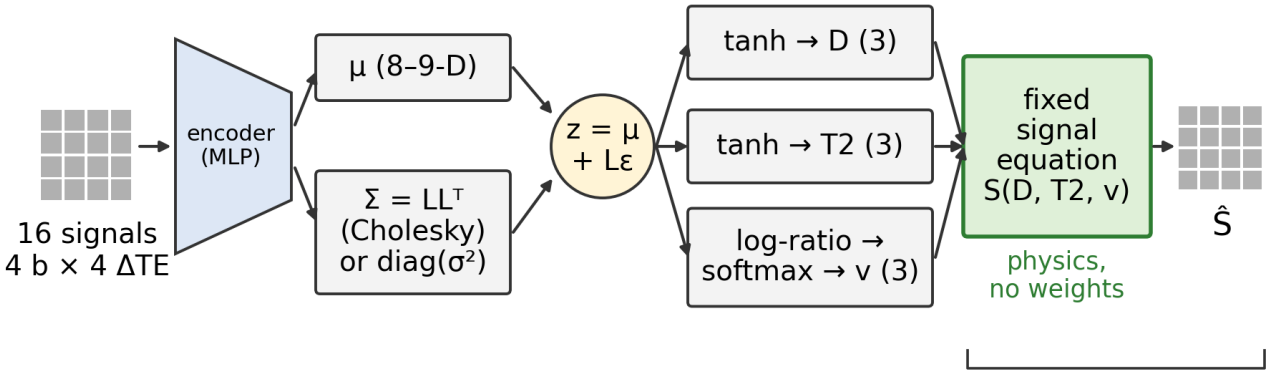
The original posterior was overconfident mainly because its implied noise SD was 56 times too small, a known VAE pitfall[11]. Under this protocol at SNR 20, compartmental D and T2 are practically non-identifiable, and every

estimator did worse than a constant, which pooled metrics hide. Limitation: data simulated with the decoder's own forward model (a best case). ROI-level coverage can be tested in vivo against registered histology fractions[3].

Conclusion

At SNR 20 on this protocol, HM-MRI fitters should report tissue fractions, not compartmental D or T2. Variational fraction intervals need a noise-matched likelihood and coverage checks across SNR and composition.

Figures



prior box

(centre ± half-width)

	D (μm²/ms)	T2 (ms)
ep	0.5 ± 0.2	45 ± 25
st	1.2 ± 0.5	70 ± 30
lum	2.85 ± 0.15	750 ± 250

loss = −log p(x | θ) + KL[q ‖ N(0, I)]

original: MSE + 0.1·KL ⇒ implied noise SD 0.9

matched: Gaussian SD = noise SD (50, or per voxel)

true noise SD 10–80 (signal units)

Figure 1: PIVA. An encoder maps 16 signals (4 b-values × 4 ΔTE) to a Gaussian posterior over latent z, diagonal (9-D) or Cholesky-factored (8-D). D and T2 are tanh-bounded to the prior box (table, centre ± half-width); fractions come from a softmax, via log-ratio coordinates for CholeskyILR. The decoder is the fixed signal equation. The original loss implies a noise SD of 0.9 signal units against a true 10–80; the matched loss uses 50 or each voxel's own noise SD.

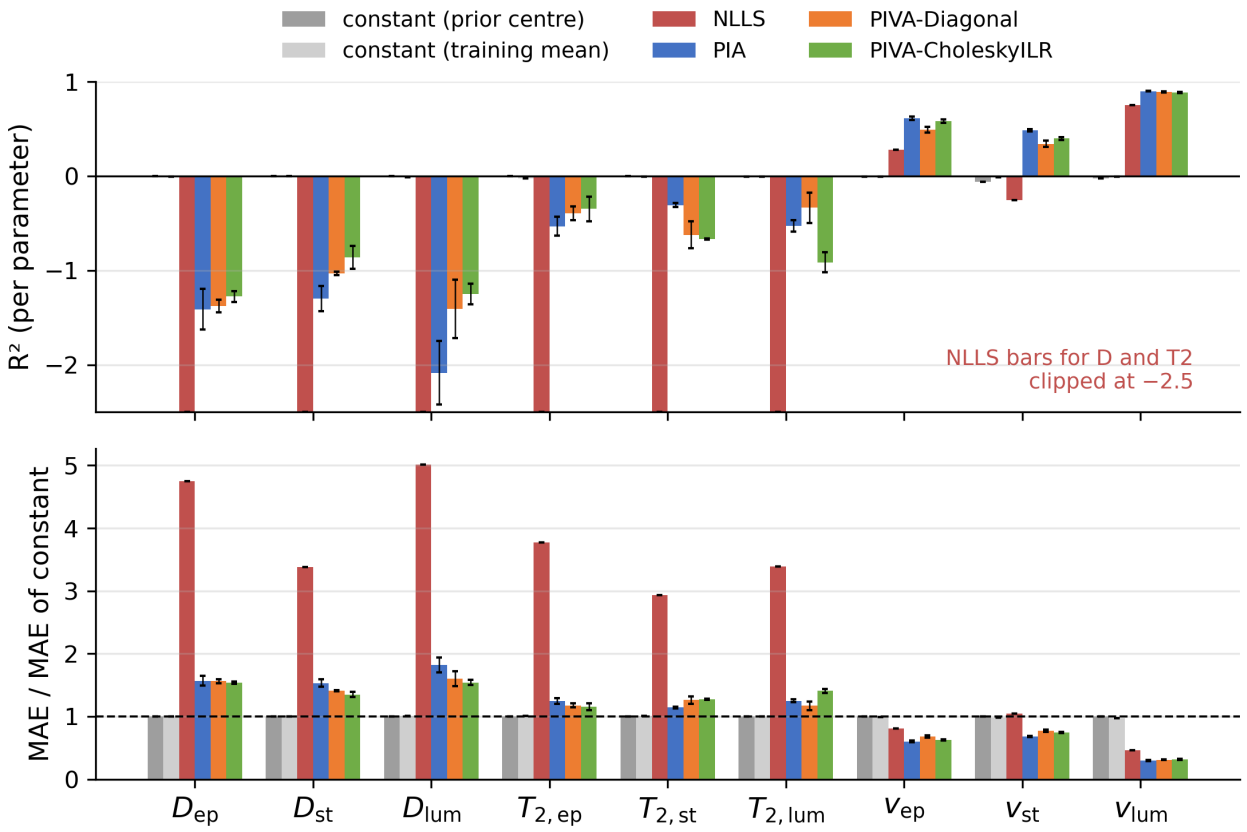


Figure 2: Per-compartment accuracy, 2000 test voxels, σ = 0.05, mean ± SD over three seeds. Top: R^2 (NLLS D and T2 bars clipped at −2.5). Bottom: MAE relative to a constant predictor at the prior centre (dashed line); grey bars are this constant and the training mean. On all six D and T2 parameters every method is worse than the constant (networks 1.15–1.8×, NLLS 2.9–5.0×); only the fractions are estimated.

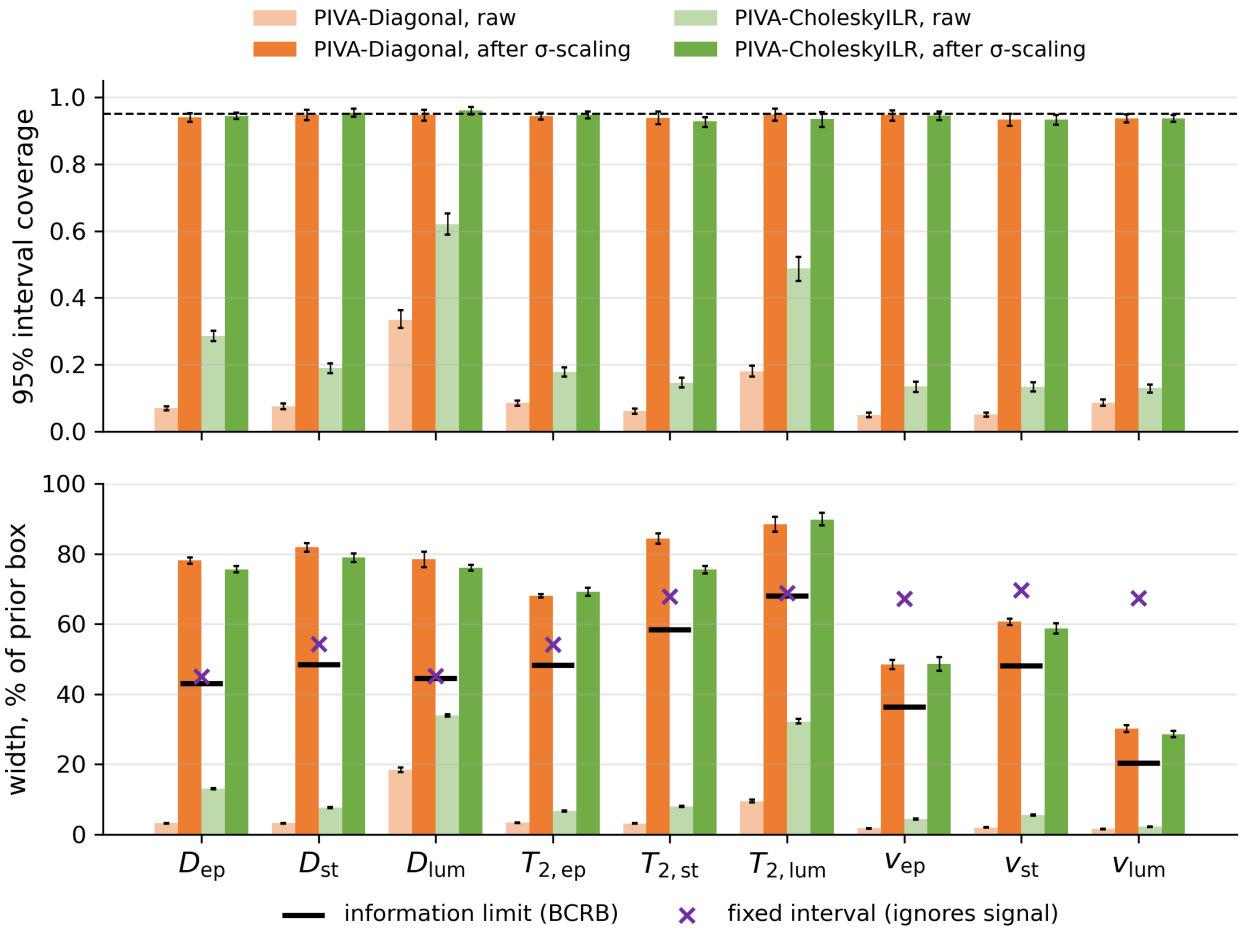


Figure 3: Original loss. Top: 95% coverage, raw (light) and after σ-scaling fit on a separate cohort (dark). Bottom: width as % of the prior box, clipped, with the information limit (BCRB, bar) and a fixed interval that ignores the signal (×). Error bars: patient-bootstrap 95% CIs. Raw coverage is 0.05–0.62. After scaling it is 0.93–0.96, but every D and T2 interval (68–90%) is wider than the fixed interval (45–69%); only fraction intervals are narrower.

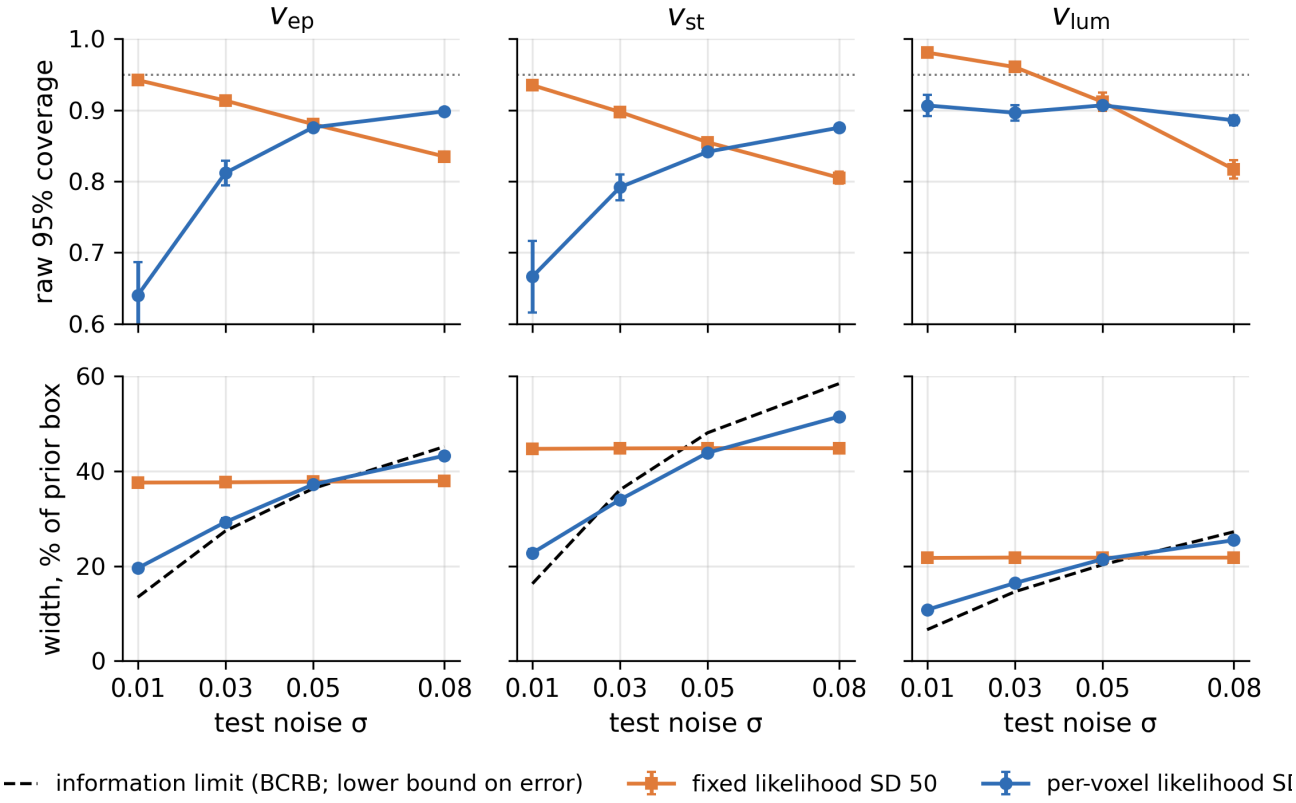


Figure 4: Fraction posteriors against test noise σ (SNR 100 to 12.5), CholeskyILR, no post-hoc scaling, mean ± SD over three seeds. Top: raw 95% coverage (dotted, 0.95). Bottom: width as % of the prior box; dashed, information limit (BCRB), a lower bound on error rather than a target width. A fixed likelihood SD of 50 gives flat widths, 2.7–3.3 times the limit at σ = 0.01, that under-cover at σ = 0.08. A per-voxel SD follows the limit for $\sigma \geq 0.03$ but under-covers at σ = 0.01.

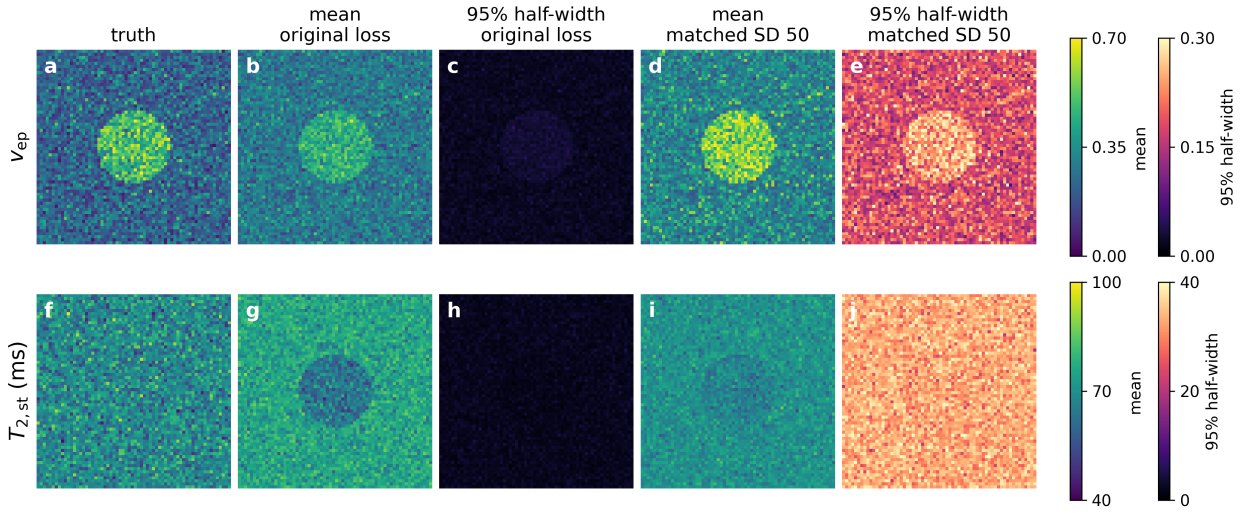


Figure 5: Synthetic phantom (σ = 0.05): a cancer-like disc (epithelium signal fraction 0.48) in benign-like tissue (0.23); stromal T2 has no disc. Top: epithelium truth, means and 95% half-widths. The original loss keeps 62% of the contrast with near-zero half-widths. The matched loss (SD 50) keeps 92% but biases both regions upward (+0.09, +0.07), with half-widths of 0.18–0.25. Bottom: the original loss paints a false −11 ms stromal T2 disc; matched intervals span the prior box.

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